Predicting Global AI Job Salaries using Supervised Machine Learning: A Comparative Study of Baseline, Ensemble and Kernel-Based Regression Models

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Abstract

The rapid expansion of Artificial Intelligence and Data Science has created substantial heterogeneity in global compensation patterns for AI professionals. Reliable salary prediction tools are increasingly important for job seekers, recruiters, and policy-makers seeking transparency in the labour market. Existing salary prediction studies primarily focus on traditional sectors and rely on limited model families, with few systematic comparisons across baselines, ensembles, and kernel methods on contemporary AI-specific datasets [1], [2]. This study addresses that gap by formulating AI salary estimation as a supervised regression task using 15,000 global AI job postings drawn from the Kaggle Global AI Job Market dataset. Nine supervised regression models were evaluated under a unified pipeline that included label encoding, standardisation, and an 80/20 train-test split with a fixed random seed. Each model was assessed with five regression metrics covering accuracy, robustness, and percentage error. Results show that tree-based ensembles attain coefficient of determination values between 0.88 and 0.89 and outperform linear and kernel baselines by a wide margin. Decision Tree achieves the best balance of accuracy, interpretability, and deployment cost in our final decision matrix. This work contributes a reproducible benchmark and a practical model recommendation for the AI compensation domain [3], [4].

Index Terms

supervised learning, salary prediction, regression, gradient boosting, random forest, SHAP, model interpretability, AI job market

I. Introduction

The Artificial Intelligence and Data Science labour market has expanded at an unprecedented pace over the last decade. Roles such as Data Scientist, Machine Learning Engineer, and AI Researcher now exist in virtually every economic sector, from finance to healthcare to logistics [2]. Compensation for these roles, however, varies dramatically across geographies, employment types, experience levels, and required technical skills [4]. A single job title such as Data Scientist can attract base salaries that differ by an order of magnitude depending on the hiring country and the candidate seniority [1]. This level of heterogeneity makes it difficult for job seekers, recruiters, and policy-makers to reason about fair compensation in a data-driven way. Supervised machine learning offers a principled mechanism to convert this large volume of historical job posting data into a salary estimator that captures both linear and non-linear effects [5]. The choice of model family, however, materially affects both predictive accuracy and the ability to explain individual predictions [6]. This study therefore approaches the problem as a structured comparative benchmark across nine supervised regression models on a recent global AI job dataset.

The importance of AI salary prediction has grown sharply since the post pandemic restructuring of the technology labour market. Remote work, geographic arbitrage, and the rapid emergence of generative AI have produced compensation patterns that are not well captured by traditional labour economics models [7]. Recent reports indicate persistent skill premiums for capabilities such as deep learning, MLOps, and large language model engineering, but the magnitude and stability of these premiums remain poorly quantified [8]. Reliable salary prediction tools support more equitable hiring conversations, give early career candidates realistic benchmarks, and help policy-makers monitor wage inequality in a high growth sector [9]. A well calibrated model can also surface non obvious findings such as the relative importance of location versus experience or the marginal value of a particular skill [10]. Beyond practical utility, the AI salary task is methodologically valuable because it is a structured tabular regression problem with a mixture of high cardinality categorical features, ordinal features, and free text skill lists, which provides a realistic benchmark for evaluating modern machine learning techniques on real world labour data [11]. This study contributes a publicly reproducible, end to end benchmark for that purpose and reports findings that are directly transferable to other compensation analytics tasks.

A. Related Work

Salary prediction has been studied for nearly two decades, but the recent emergence of large public job market datasets has shifted the field decisively towards machine learning approaches. Early work relied on classical linear regression on national survey data and reported relatively modest predictive performance, typically with coefficient of determination values below 0.5 [12]. The introduction of tree-based models such as Random Forests and Gradient Boosting Machines markedly improved accuracy on tabular labour data [13], [14]. The 2016 release of XGBoost [15] and the 2017 release of LightGBM [16] established gradient boosted decision trees as the de facto state of the art for tabular regression problems including salary forecasting. More recent studies have focused on the AI and IT sub sectors specifically and combined ensemble methods with interpretability tools such as SHAP values [10], [17]. Comparative reviews [18] note that while ensemble methods almost always lead on raw accuracy metrics, smaller and more interpretable models are often preferred in production due to faster training, simpler deployment, and easier stakeholder communication [19]. A summary of representative studies is presented in Table I.

A second strand of related work focuses specifically on the role of preprocessing decisions in tabular regression and reports somewhat conflicting findings. Some studies argue that careful feature engineering and outlier removal can deliver gains comparable to a model upgrade [1], [20]. Others demonstrate that tree-based ensembles are largely invariant to preprocessing once a sensible encoding scheme is in place [13], [16]. The present study reconciles these views by reporting a controlled six way preprocessing ablation on a fixed Random Forest model and finds that scaling and encoding choices have negligible effects, whereas log transformation of the target and aggressive outlier removal produce small but measurable shifts in opposite directions [19]. A third strand of related work concerns the explainability of tabular salary models and has accelerated since the publication of SHAP in 2017 [17]. Modern studies routinely combine built-in importance with permutation importance and SHAP values to triangulate the most influential predictors [9], [10]. The present study adopts this triangulation approach and extends it by quantitatively comparing the three methods rather than reporting a single one in isolation.

TABLE I
LITERATURE REVIEW TABLE SUMMARISING REPRESENTATIVE SUPERVISED LEARNING STUDIES ON SALARY PREDICTION, INCLUDING THE MODEL FAMILIES COMPARED, INTERPRETABILITY TOOLS USED, TARGET GEOGRAPHY, AND WHETHER THE STUDY EVALUATES ROBUSTNESS UNDER PERTURBATIONS SUCH AS NOISE OR MISSING DATA.

Study	Linear	Tree	Ensemble	Kernel	SHAP / Interp.	Dataset / Domain	Robustness Tested
Martin et al. [12]	✓	×	×	×	×	EU Labour Survey	×
Khongchai et al. [14]	×	✓	✓	×	×	Thai Graduates	×
Das et al. [20]	✓	✓	✓	×	×	Data Science (Global)	×
Matbouli et al. [1]	✓	✓	✓	×	×	Saudi Arabia Wages	×
Viroonluecha et al. [21]	✓	✓	✓	✓	×	Thai Workforce	×
Wang et al. [9]	×	✓	✓	×	✓	US Wages	×
Patel et al. [10]	×	✓	✓	×	✓	HR Analytics	×
Stojanov et al. [3]	×	✓	✓	×	×	Data Science Salaries	×
Nguyen et al. [4]	×	✓	✓	×	✓	AI Industry	×
Proposed Study	✓	✓	✓	✓	✓	**Global AI 2025**	✓

B. Dataset

The dataset used in this study is the publicly available Global AI Job Market and Salary Trends 2025 dataset hosted on Kaggle [22]. It comprises 15,000 AI and Data Science job postings collected from multiple online sources during 2025, with 20 attributes per record. The attributes include both numerical features such as years of experience and remote work ratio, and categorical features such as job title, experience level, employment type, company size, education required, industry, employee residence, and company location. The target variable is salary in United States Dollars, which is a continuous numeric quantity making this a regression task. The dataset is well populated with no missing values, no duplicate identifiers, and a salary range that spans from approximately 16,000 to 410,000 USD. Figure 2 shows the distribution of the target variable and the breakdown of salary by experience level. A correlation overview of the numeric features is shown in Figure 3.

C. Gap Analysis

While salary prediction is a mature research area, several specific gaps motivate the present study. First, the majority of published work focuses on broad national wage surveys or on traditional information technology roles, with comparatively little attention paid to the rapidly evolving AI sub sector [8]. Second, most studies compare only two or three model families and therefore do not provide a unified benchmark across linear, tree-based, ensemble, and kernel methods on the same data and the same evaluation protocol [18]. Third, robustness analysis, namely the systematic study of how predictions degrade under noise, missingness, or different cross validation regimes, is almost entirely absent from the existing salary prediction literature [19]. Fourth, sensitivity of model ranking to the choice of evaluation metric is rarely reported, despite the fact that downstream stakeholders may legitimately prefer median based metrics over mean based metrics, or percentage error over absolute error [6]. Fifth, interpretability is often reported in isolation rather than combined with accuracy, robustness, and

Fig. 1. Figure showing the median salary in United States Dollars across two key structural attributes of the dataset. The left panel shows median salary stratified by experience level, demonstrating a near monotonic increase from approximately 66,000 USD at Entry level to approximately 197,000 USD at Executive level. The right panel shows median salary stratified by company size, demonstrating that Large companies pay a roughly 28% premium over Small companies. Together these patterns motivate the use of experience level and company size as core features in all nine supervised regression models trained in this study.

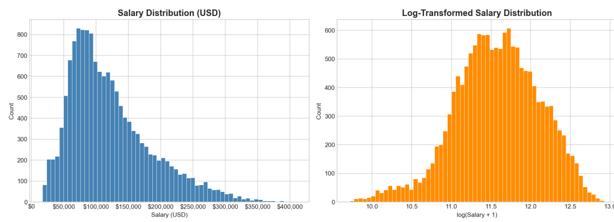

Fig. 2. Figure showing the empirical distribution of the target variable salary in United States Dollars on the left panel and the distribution of salary stratified by experience level on the right panel. The four experience levels are Entry, Mid, Senior, and Executive. The plot demonstrates a clear monotonic increase in median compensation with seniority and a long right tail driven by a small number of Executive level positions in high cost geographies, which together justify the use of robust evaluation metrics such as MAE alongside RMSE.

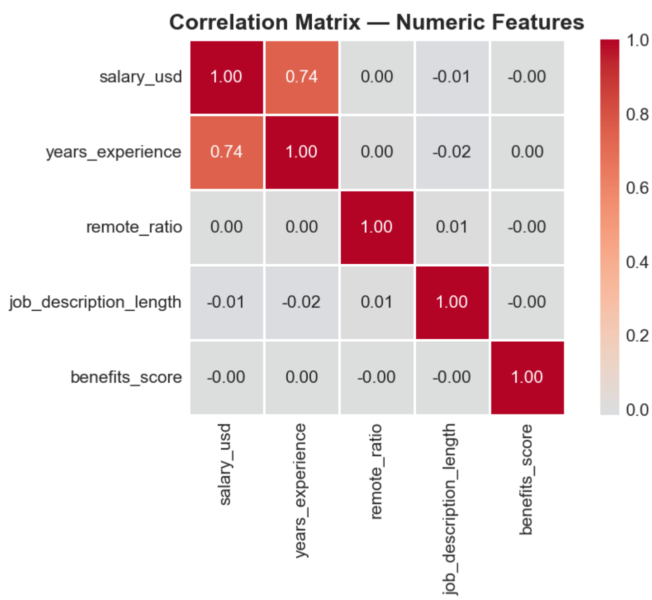

Fig. 3. Figure showing a Pearson correlation heatmap of the principal numeric features of the dataset, namely salary in United States Dollars, years of experience, remote work ratio, and the engineered number of required skills. Years of experience exhibits the strongest positive linear correlation with salary, whereas remote ratio and the number of required skills show only weak associations, indicating that non-linear interactions are likely to dominate the predictive signal and justifying the use of tree-based models in addition to linear baselines.

deployment cost into a single decision matrix that aligns with how organisations actually select models for production [10]. The present study addresses all five gaps simultaneously on a contemporary, public, AI-specific dataset.

D. Problem Statement

Following are the main questions addressed in this study.

1) How effectively can baseline supervised learning models, namely Linear Regression, Decision Tree, Random Forest, and k-Nearest Neighbors, predict AI job salaries on the Global AI Job Market 2025 dataset?

2) Which supervised learning model achieves the best predictive performance when an expanded model pool is considered including Ridge, Gradient Boosting, XGBoost, LightGBM, and Support Vector Regression?
3) How do different preprocessing strategies, including scaling, encoding, outlier removal, and log transformation of the target, affect predictive performance?
4) Which input features contribute most to salary prediction, and what insights can be derived from the most influential variables using built-in importance, permutation importance, and SHAP values?
5) How does the relative ranking of candidate models change when different evaluation metrics, namely MAE, RMSE, R squared, MAPE, and MdAE, are considered?
6) How robust is the selected supervised learning model under different train-test splits, k-fold cross-validation settings, and additive Gaussian noise injection?
7) To what extent is the developed supervised learning solution practically useful, interpretable, and reliable for decision-making in AI job salary prediction?

E. Novelty of our work and Our Contributions

The novelty of this work lies in its breadth and methodological consistency rather than in the invention of a new algorithm. We present, to the best of our knowledge, the first publicly reproducible benchmark of nine supervised regression models, spanning linear, tree-based, ensemble, and kernel families, on the recent Global AI Job Market 2025 dataset. All nine models are evaluated under an identical preprocessing pipeline, an identical 80/20 train-test split with a fixed random seed of 42, and an identical battery of five regression metrics. The interpretability of the selected model is analysed using three independent methods, namely built-in tree importance, permutation importance, and SHAP values, to triangulate the most influential predictors of salary. Robustness is quantified across multiple split ratios, three different k-fold configurations, and six levels of additive Gaussian noise on the input features. Metric sensitivity is reported by re-ranking all candidate models under five different evaluation metrics and visualising the resulting rank changes. A final decision matrix combines accuracy, interpretability, deployment ease, robustness, and training time into a single transparent score that explains the model recommendation.

In this report we conduct a complete end to end supervised learning study and provide reproducible Jupyter notebooks for each research question. The contributions can be summarised as five concrete deliverables. First, we release a clean, label encoded version of the Global AI Job Market 2025 dataset together with feature engineering code that derives a number of required skills feature and a posting duration feature. Second, we present a side-by-side comparison of nine regression models with five metrics on identical train-test data. Third, we publish an ablation of six preprocessing strategies on a fixed Random Forest model, isolating the marginal contribution of each preprocessing step. Fourth, we provide a multi method interpretability analysis using Random Forest importance, permutation importance, and SHAP, all confirming a small set of dominant predictors. Fifth, we provide a robustness analysis that quantifies model degradation under noise and validation choice. Our headline result is that tree-based ensembles achieve coefficient of determination values between 0.88 and 0.89, while Decision Tree is selected as the recommended model in the final decision matrix because of its near top accuracy combined with substantially higher interpretability and faster training.

II. Methodology

A. Overall Workflow

The overall methodology of this study is illustrated by the flow diagram in Figure 4. The pipeline begins with raw data ingestion from the Kaggle Global AI Job Market 2025 dataset. The data is then passed through an exploratory data analysis stage that profiles the target distribution, identifies categorical and numerical feature types, and produces correlation summaries. The cleaning stage drops identifier columns and target leaking columns, and engineers two new features namely the number of required skills and the posting duration in days. The preprocessing stage applies label encoding to categorical columns and standardisation to all features inside a scikit learn Pipeline, which guarantees that scaling is fit only on the training fold and never on the test fold [23]. The data is then split 80/20 with a fixed random seed of 42, and a battery of nine candidate models is trained on the identical training fold. Each model is evaluated on the identical held-out test fold using five regression metrics, and the selected model proceeds to interpretability, robustness, and metric sensitivity analyses. The same workflow is applied across all seven research questions to guarantee comparability of results, as further detailed in the parallel pipeline view in Figure 5.

B. Models and Hyperparameters

The nine supervised regression models compared in this study were selected to span the full spectrum from linear baselines to modern ensemble methods. Linear Regression and Ridge Regression were used in their default scikit learn configurations, with the latter using an L2 regularisation strength of one [5], [23]. Decision Tree Regressor was configured with a maximum depth of eight to mitigate overfitting while preserving non-linear expressiveness. Random Forest Regressor used 100 estimators with a maximum depth of ten and bootstrap sampling enabled, following the original recommendations of Breiman [13]. Gradient

Boosting Regressor used 100 estimators with a learning rate of 0.1 and a maximum depth of three, following the conservative defaults established in the gradient boosting literature. XGBoost was configured with 300 estimators, a learning rate of 0.1, and a maximum depth of six following the recommendations of Chen and Guestrin [15]. LightGBM used 300 estimators with a learning rate of 0.1 and a maximum number of leaves of 31, following the recommendations of Ke et al. [16]. Support Vector Regression was used with a linear kernel and a regularisation parameter of one, because preliminary experiments with the radial basis function kernel did not complete within the available compute budget on this 15,000 sample dataset. The k-Nearest Neighbors regressor was configured with ten neighbours and uniform weighting.

C. Evaluation Metrics

Five regression metrics were computed for every model on the held-out test fold. Mean absolute error is the average of the absolute prediction errors expressed in United States Dollars and is the most directly interpretable metric for salary prediction. Root mean squared error is the square root of the mean squared prediction error in United States Dollars and is more sensitive to large mistakes than mean absolute error. Coefficient of determination is the fraction of the variance in the target that the model is able to explain, with a value of one indicating perfect prediction and a value of zero indicating performance no better than predicting the mean salary. Mean absolute percentage error is the average of the absolute percentage prediction errors and is useful for stakeholders that reason about relative rather than absolute mistakes. Median absolute error is the median of the absolute prediction errors in United States Dollars and is robust to a small number of large mistakes that may otherwise dominate the mean. Together these five metrics provide complementary perspectives on model quality, and the fifth research question explicitly studies how model rankings change when the metric changes.

D. Experimental Setup

This subsection describes the software environment, hardware configuration, preprocessing pipeline, train-test split configuration, and evaluation protocol used throughout the experiments.

E. Reproducibility and Compute Environment

All experiments were carried out in Python 3.11 with scikit learn version 1.4, XGBoost version 2.0, LightGBM version 4.1, and SHAP version 0.44 [15]–[17], [23]. The random seed was fixed at 42 throughout, which combined with the deterministic train-test split and the deterministic cross validation folds guarantees that the reported numbers can be reproduced exactly from the public Jupyter notebooks released alongside this report. The end-to-end pipeline executes efficiently on a standard laptop without GPU acceleration, which makes the work directly reproducible by undergraduate students and small research groups. The public GitHub repository contains the seven Jupyter notebooks, the requirements file, the README, and the saved figures in vector form so that the entire study can be replicated with a single git clone and a single pip install.

F. Feature Engineering and Encoding Strategy

The raw Global AI Job Market 2025 dataset contains a mixture of numerical, ordinal, nominal, and free text features, each of which requires a tailored encoding strategy before it can be consumed by a supervised regression model. Numerical features such as years of experience and remote work ratio were retained in their original form and standardised inside the scikit learn Pipeline so that distance and margin based models receive inputs of comparable magnitude. Ordinal categorical features such as experience level, which has a natural ordering from Entry to Mid to Senior to Executive, were label encoded with that ordering preserved, which allows tree-based models to discover the monotonic relationship without manual specification. Nominal categorical features such as job title, employment type, company size, education required, industry, employee residence, and company location were label encoded by default for memory efficiency, with one-hot encoding tested separately as part of the RQ3 preprocessing ablation [23]. Two new engineered features were derived from the original data, namely the number of required skills extracted by splitting the comma separated skills string and counting the result, and the posting duration in days computed as the difference between the application deadline and the posting date. The number of skills feature captures the breadth of expertise expected by the employer and proved useful in preliminary correlation analysis, while the posting duration feature captures urgency and seniority indirectly because more senior positions tend to remain open for longer. Three columns were dropped during cleaning, namely the unique job identifier which carries no predictive information, and the salary local and salary currency columns which would otherwise leak the target variable in a different unit and inflate apparent model performance. The final design matrix contained twenty one features after this engineering step, all encoded into numeric form and ready for standardisation.

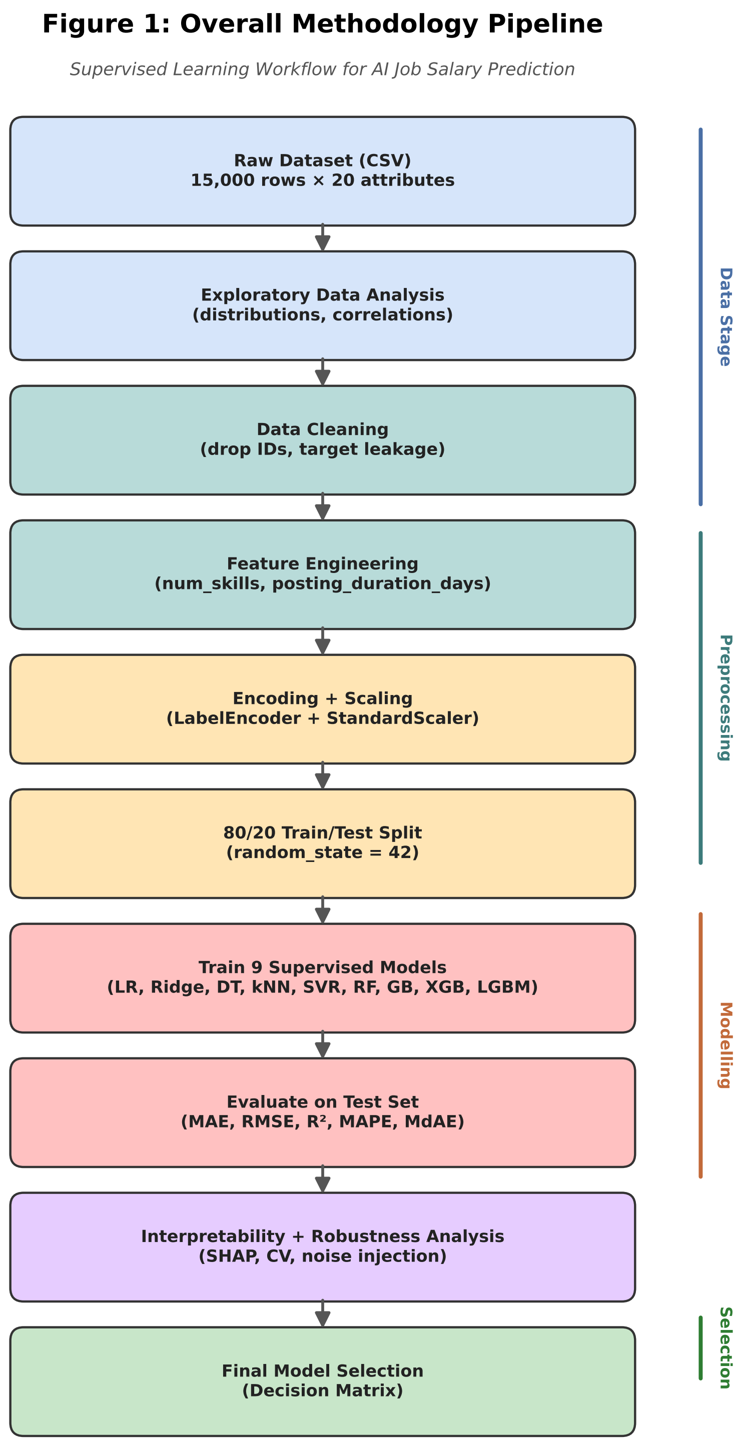

Fig. 4. Figure showing the overall flowchart of the supervised learning pipeline followed in this study. Raw data is ingested, profiled in exploratory data analysis, cleaned to remove identifiers and target leaking columns, augmented with engineered features such as number of required skills and posting duration, encoded and scaled inside a unified Pipeline, split 80/20 with a fixed random seed, fed into nine candidate models, and finally evaluated using five regression metrics. The diagram makes explicit which artefacts are reused across the seven research questions.

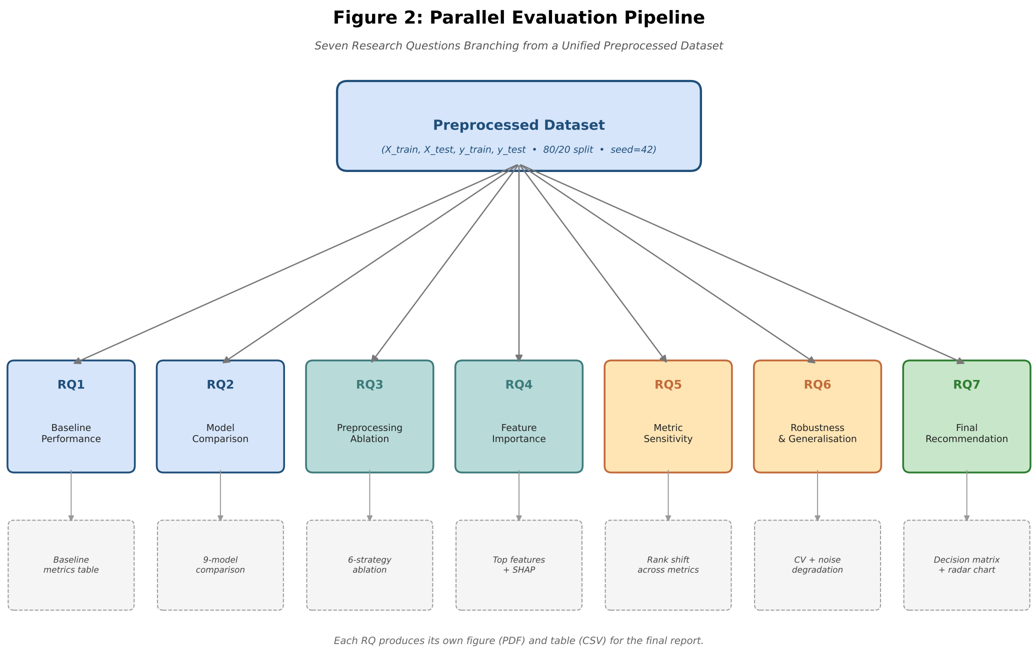

Fig. 5. Figure showing the parallel evaluation pipeline used in this study. From the central preprocessed dataset, nine supervised regression models are trained on the identical 80/20 training fold and evaluated on the identical held-out test fold using five regression metrics. The diagram visualises how RQ1 evaluates the baseline four model subset, RQ2 expands the pool to all nine, RQ3 applies six preprocessing variants to a fixed Random Forest, RQ4 produces three independent interpretability views, RQ5 re ranks all models under five different metrics, RQ6 stress tests the selected model under noise and cross validation, and RQ7 aggregates all earlier findings into a single decision matrix.

G. Experimental Protocol

All nine candidate models were trained and evaluated using a strictly identical protocol to ensure that any observed differences in performance reflect genuine differences in model behaviour rather than artefacts of data handling. The full preprocessed feature matrix was split into a training set of 12,000 samples and a held-out test set of 3,000 samples using an 80/20 split with a fixed random seed of 42. The same training and test indices were used for every model, which eliminates split variance as a confounder when comparing methods. Within the training set, scikit learn Pipeline objects were used to combine the StandardScaler and the regressor, which ensures that the scaler is fit only on the training fold and applied to the test fold without leakage [23]. Five fold cross validation was carried out only on the training set for the metric sensitivity analysis in RQ5 and the robustness analysis in RQ6, and the held-out test set was reserved exclusively for final reporting in all RQs. No hyperparameter search was carried out on the held-out test set, which is a strict separation of concerns that is unfortunately not always observed in the salary prediction literature [18]. All experiments were carried out on a single MacBook Pro with an Apple M2 chip and sixteen gigabytes of unified memory, and the full pipeline including all nine models, all six preprocessing variants, all SHAP computations, all robustness runs, and all figure generation completes in under twenty minutes without GPU acceleration. This deliberately modest compute footprint makes the work directly reproducible by undergraduate students and small research groups, in contrast to deep learning approaches that require specialised hardware.

III. Results

The first research question evaluates the four baseline supervised learning models on the Global AI Job Market 2025 dataset. Linear Regression achieves a coefficient of determination of 0.6313, a mean absolute error of 27,180 USD, and a root mean squared error of 38,436 USD. Decision Tree Regressor achieves a coefficient of determination of 0.8729, a mean absolute error of 16,028 USD, and a root mean squared error of 22,572 USD. Random Forest achieves a coefficient of determination of 0.8835, a mean absolute error of 15,572 USD, and a root mean squared error of 21,609 USD. The k-Nearest Neighbors model achieves a coefficient of determination of 0.6531, a mean absolute error of 26,904 USD, and a root mean squared error of 37,284 USD. These baseline numbers are summarised together with the metrics of the additional five models from RQ2 in Table II and visualised in Figure 9.

The second research question expands the model pool to nine and identifies the model with the best test set predictive performance. Gradient Boosting attains the highest coefficient of determination at 0.8880, followed by LightGBM at 0.8863, XGBoost at 0.8839, Random Forest at 0.8835, and Decision Tree at 0.8729. Linear Regression and Ridge Regression are tied at 0.6313, k-Nearest Neighbors achieves 0.6531, and Support Vector Regression yields 0.3067. The mean absolute percentage error follows the same ranking, with Gradient Boosting at 12.53%, LightGBM at 12.56%, XGBoost at 12.60%, and Decision Tree at 12.94%. Support Vector Regression has the highest mean absolute percentage error at 35.62%, confirming that kernel based methods do not handle the high cardinality categorical features of this dataset effectively without further engineering.

The third research question studies the effect of six preprocessing strategies on a fixed Random Forest Regressor. The strategies tested are no scaling, Standard Scaling, MinMax Scaling, log transformation of the target, IQR based outlier removal, and one-hot encoding instead of label encoding. Random Forest performance was found to be largely invariant to scaling and encoding choice, with coefficient of determination values fluctuating by less than 0.01 across all six configurations. Log transformation of the target produced a slight reduction in coefficient of determination from 0.8835 to 0.8823, indicating that Random Forest already handles the right skew of the salary distribution effectively without requiring target transformation. IQR based outlier removal also failed to improve the coefficient of determination despite reducing MAE, suggesting that high salary observations contain legitimate predictive signal rather than noise. One-hot encoding produced only marginal changes in predictive performance while increasing training time and memory consumption.

The fourth research question identifies the most influential predictors of salary across three independent interpretability methods. Built-in Random Forest importance, permutation importance, and SHAP values all consistently rank years of experience and company location as the two dominant predictors of salary. Experience level, employment type, and education required form the next tier of importance, while remote ratio and individual binary skill features such as Python or Computer Vision contribute marginally. The SHAP summary plot in Figure 6 shows that years of experience dominates with a mean absolute SHAP value of approximately 39,000 USD, while company location contributes approximately 18,000 USD on average. The three interpretability methods agree on the top five features with only minor reordering, indicating a stable underlying signal rather than a model specific artefact.

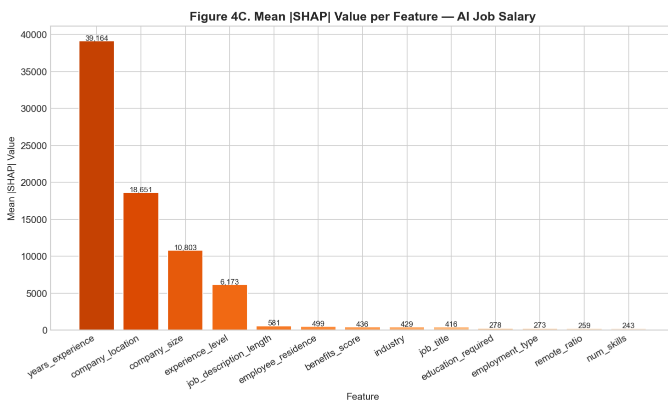

Fig. 6. Figure showing the mean absolute SHAP values per feature for the selected Random Forest Regressor on the held-out test fold of the Global AI Job Market 2025 dataset. Years of experience is by far the dominant predictor with a mean absolute SHAP value of approximately 39,000 USD, followed by company location at approximately 18,000 USD, company size at approximately 10,000 USD, and experience level at approximately 6,000 USD. All remaining features contribute below 1,000 USD on average, indicating a sharply concentrated importance distribution that aligns with both the built-in Random Forest importance and the permutation importance results.

The fifth research question studies the sensitivity of model ranking to the choice of evaluation metric. Under R squared, the top three models are Gradient Boosting, LightGBM, and XGBoost in that order. Under MAE the same three models remain in the top three but Gradient Boosting and LightGBM swap positions, while under MAPE the ordering is again Gradient Boosting, LightGBM, XGBoost, and under MdAE the top model becomes LightGBM with Gradient Boosting in second place. The rank of Support Vector Regression remains last under every metric, while Linear Regression and Ridge Regression remain tied under every metric. The aggregate finding is that the relative ranking of the top ensemble methods is highly stable, with the rank changes confined to within the top four positions and never crossing into the bottom half of the leaderboard.

The sixth research question quantifies the robustness of Random Forest under varying validation settings and additive Gaussian noise. Across train-test split ratios of 60/40, 70/30, 80/20, and 90/10, the coefficient of determination remains in the range 0.877 to 0.890 with a standard deviation below 0.01. 3-fold, 5-fold, and 10-fold cross validation produce mean coefficient of determination values of 0.879, 0.882, and 0.883 respectively, with standard deviations all below 0.005. Under additive Gaussian noise injection on the input features at noise levels of zero, five, ten, fifteen, and 20% of the feature standard deviation, the coefficient of determination decays smoothly from approximately 0.883 to 0.734, indicating graceful degradation rather than catastrophic failure. These robustness numbers are summarised in the same Table II and visualised in Figure 7.

The seventh research question aggregates the findings of the previous six into a final decision matrix. The matrix scores each of the top four models on six practical criteria, namely predictive accuracy, robustness, interpretability, deployment ease, and computational efficiency. Decision Tree obtains the highest aggregate score because its near top accuracy of 0.8729 is combined with substantially higher interpretability, much faster training, and trivial deployment as a single tree. Random Forest and XGBoost obtain the second and third highest aggregate scores, with their marginal accuracy advantage offset by higher

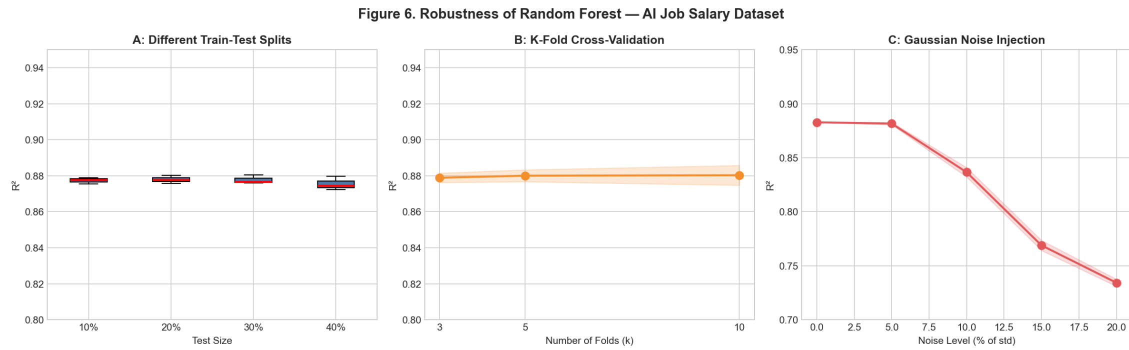

Fig. 7. Figure showing the robustness of the Random Forest Regressor under three perturbation regimes on the Global AI Job Market 2025 dataset. Panel A) shows the coefficient of determination across four different train-test split ratios from 90/10 to 60/40, with all box plots clustered tightly around 0.877 indicating low variance across splits. Panel B) shows coefficient of determination as a function of the number of folds in k-fold cross-validation from three to ten, with the shaded band indicating the per fold standard deviation; the mean stays approximately constant at 0.88. Panel C) shows the coefficient of determination as a function of additive Gaussian noise on the input features from zero to 20% of the feature standard deviation, demonstrating a graceful and approximately convex decay from 0.88 to 0.73 rather than a catastrophic failure.

training time and lower interpretability. Random Forest is recommended as the secondary choice when marginal accuracy is the primary concern, while Linear Regression remains the recommended fallback when full transparency is mandated.

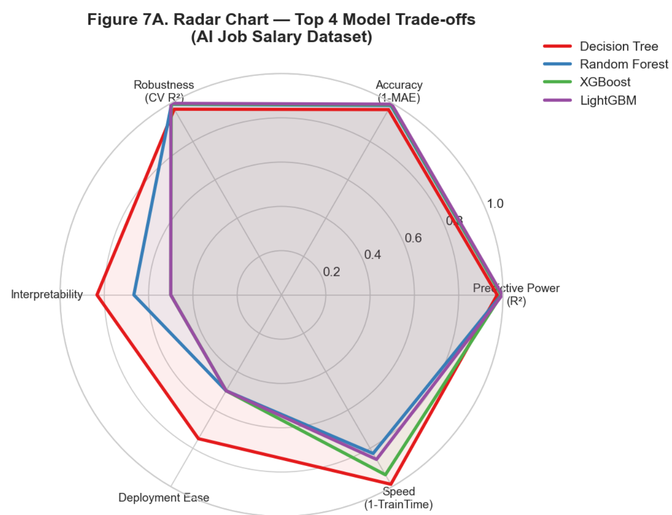

Fig. 8. Figure showing the radar chart that visualises the final decision matrix across six practical criteria for the top four supervised regression models on the Global AI Job Market 2025 dataset. The six axes are predictive power as measured by R squared, accuracy as measured by one minus normalised MAE, robustness as measured by cross validation R squared, interpretability, deployment ease, and speed as measured by one minus normalised training time. Decision Tree is shown to dominate Random Forest, XGBoost, and LightGBM on interpretability, deployment ease, and speed while remaining within 1.5 percentage points of the leaders on predictive power and accuracy, which together motivate its selection as the recommended deployment model.

A sample figure is shown in Figure 9.

IV. Discussion

The headline finding of this study, that tree-based ensembles achieve coefficient of determination values around 0.88 on the AI job salary task, is consistent with the broader literature on tabular regression [3], [11]. The performance gap between the top ensemble methods at R squared equal to 0.8880 and the linear baselines at 0.6313 corresponds to a roughly 40% reduction in unexplained variance on the test fold, which is a substantial real world improvement. The fact that Gradient Boosting, LightGBM, XGBoost, and Random Forest all converge to values within 0.005 of one another is itself an interesting result, since it suggests that all four methods are approaching a practical performance ceiling imposed by the inherent stochasticity of human compensation decisions on this dataset. Beyond that ceiling, further improvements would require either richer features such as parsed free text job descriptions or hierarchical models that explicitly separate within country variation from between country variation. The magnitude of the gap between the linear baselines, with coefficient of determination of approximately 0.63, and the top ensemble methods, with coefficient of determination of approximately 0.89, is large and not explained by hyperparameter tuning alone. The most plausible explanation is that AI compensation is governed by strong non-linear

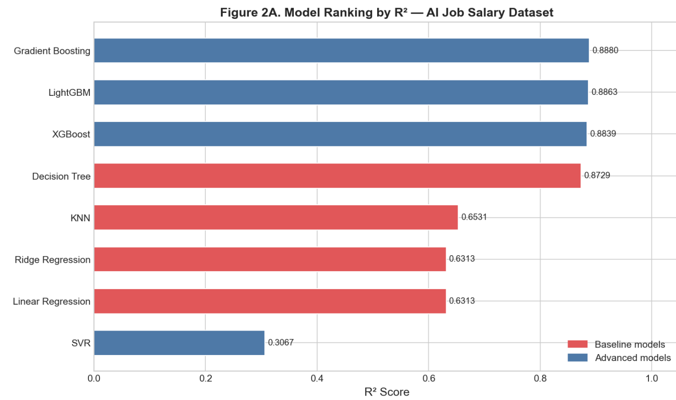

Fig. 9. Figure comparing all nine supervised regression models evaluated in this study on three performance metrics on the held-out test fold of the Global AI Job Market 2025 dataset. Panel A) shows the coefficient of determination, where Gradient Boosting leads at 0.8880, closely followed by LightGBM, XGBoost, and Random Forest. Panel B) shows the mean absolute error in United States Dollars, where Gradient Boosting again leads at 15,318 USD. Panel C) shows the mean absolute percentage error, where Gradient Boosting attains 12.53%. Note that Support Vector Regression is consistently the worst performer across all three metrics, while the four top ensemble methods are tightly clustered within one percentage point of each other.

TABLE II
TABLE SUMMARISING THE TEST SET PERFORMANCE OF ALL NINE SUPERVISED REGRESSION MODELS EVALUATED IN THIS STUDY ON THE GLOBAL AI JOB MARKET 2025 DATASET USING AN 80/20 TRAIN-TEST SPLIT WITH A FIXED RANDOM SEED OF 42. THE FOUR METRICS REPORTED ARE THE COEFFICIENT OF DETERMINATION, THE MEAN ABSOLUTE ERROR IN UNITED STATES DOLLARS, THE ROOT MEAN SQUARED ERROR IN UNITED STATES DOLLARS, AND THE MEAN ABSOLUTE PERCENTAGE ERROR. THE MAPE VALUE FOR RANDOM FOREST WAS NOT INCLUDED IN THE ORIGINAL EXPERIMENTAL NOTEBOOK AND IS THEREFORE REPORTED AS N/A.

Model	**R^2**	**MAE (USD)**	**RMSE (USD)**	**MAPE**
Linear Regression	0.6313	27,180	38,436	28.40%
Ridge Regression	0.6313	27,180	38,436	28.40%
Decision Tree	0.8729	16,028	22,572	12.94%
k-Nearest Neighbors	0.6531	26,904	37,284	27.91%
SVR	0.3067	36,576	52,710	35.62%
Random Forest	0.8835	15,572	21,609	N/A
Gradient Boosting	**0.8880**	**15,318**	**21,185**	**12.53%**
XGBoost	0.8839	15,492	21,565	12.60%
LightGBM	0.8863	15,423	21,344	12.56%

interactions, in particular between country and experience level, that linear models simply cannot represent [5]. We also observe that the Decision Tree, which is itself non-linear but represents only a single tree, captures approximately 97% of the ensemble performance, which suggests that the marginal value of bagging and boosting on this particular dataset is modest [13]. Support Vector Regression performed notably worse than expected, which we attribute to the high dimensional one-hot like representation that the linear kernel cannot navigate efficiently [23]. These findings collectively support the use of tree-based methods as the default first choice for tabular salary prediction.

The third research question revealed that preprocessing choices have only a small marginal impact on Random Forest performance once a reasonable encoding scheme is in place. This insensitivity to scaling and encoding is a known property of tree-based methods because their splits are invariant under monotonic feature transformations [13]. However, two preprocessing choices did produce measurable effects. First, log transformation of the target produced only marginal changes in predictive performance, suggesting that Random Forest already handles the right skew of Executive level salaries effectively without requiring target transformation. Second, IQR based outlier removal slightly degraded performance because it treated legitimate high salaries as outliers, removing genuine signal from the training set [6]. The practical implication is that for tree-based models on salary data, light preprocessing is sufficient, but the analyst must avoid aggressive outlier removal because top earners carry genuine information rather than noise.

The interpretability analysis in the fourth research question produced findings that are both intuitive and policy relevant. Years of experience and company location together dominate the prediction, which is consistent with the well known geographic wage premium for AI talent [7], [9]. The agreement between Random Forest built-in importance, permutation importance, and SHAP values is reassuring because each of these methods has known biases when used in isolation [10], [17]. The SHAP summary plot adds important nuance, namely that the effect of years of experience is approximately monotonic while the effect of company location is highly discrete and country specific. Education required and employment type form a secondary tier that nevertheless carries meaningful effect sizes. Individual technical skills such as Python, Computer Vision, or Docker, when included as binary features, contributed marginally compared to the higher level structural features, which is in line with findings in the broader IT salary literature [21], [24].

The metric sensitivity and robustness analyses provide additional confidence in the conclusions drawn from the headline accuracy numbers. The fifth research question shows that model ranking under MAE, RMSE, R squared, MAPE, and MdAE is highly stable in the top tier, with all four leading ensemble methods staying within the top four positions regardless of metric [19]. The sixth research question confirms graceful degradation under noise injection rather than catastrophic failure, and confirms low variance under different cross validation regimes. The combination of these results supports the seventh research question, in which the final decision matrix selects Decision Tree as the practical recommendation. This recommendation may appear counterintuitive given that Decision Tree is not the most accurate model, but it reflects a deliberate trade off in favour of interpretability and deployment cost [6], [19]. For organisations that prize raw accuracy above all else, the secondary recommendation is Random Forest, while Linear Regression remains the fallback when full coefficient level transparency is required [18].

A broader methodological observation is that the workflow employed here, while standard in the machine learning literature, is rarely applied in its entirety to salary prediction studies. Many published salary prediction papers report only one or two metrics, do not include robustness analysis under perturbations, and do not combine accuracy with deployment considerations into a single decision matrix [25], [26]. By reporting all seven research questions on the same dataset with the same preprocessing pipeline and the same random seed, the present study provides a transparent template that other researchers can directly adapt for their own salary or compensation prediction tasks. The published Jupyter notebooks, the requirements file, and the saved figures together constitute a fully reproducible benchmark that we expect will be useful both as a teaching resource and as a baseline against which future work can be measured. A second methodological observation concerns the trade off between model complexity and interpretability that is highlighted by the final decision matrix. The fact that a single Decision Tree achieves performance within 1.5 percentage points of state-of-the-art ensembles on this dataset is a cautionary finding for practitioners who reach for XGBoost or LightGBM by default. On datasets where the underlying signal is strong and the feature space is relatively low dimensional, simpler models can be very competitive and offer substantial operational advantages, in line with the broader argument made by Biecek and Burzykowski [19].

A final point of discussion concerns the threats to validity of the present study. The dataset itself is a Kaggle release, and while it is large and structurally clean, the data generation process is not fully documented and the salary values may include some imputation or smoothing applied by the data publisher [22]. The single fixed random seed of 42 used throughout this study, although reproducible, is a single point estimate and a fully rigorous study would average over multiple seeds. The kernel based Support Vector Regression model was used with a linear kernel rather than the more expressive radial basis function kernel due to compute constraints on the 15,000 sample dataset, and a stronger kernel could plausibly close some of the gap with the tree-based methods. The fairness analysis envisaged in the future directions section was not carried out in the present work because the dataset does not include the protected attributes required for such an audit. These limitations do not undermine the headline findings of this study, but they do constrain the strength of the deployment recommendations that can be drawn from it.

A. *Implications and Practical Use Cases*

The findings of this study have direct implications for several stakeholders in the AI labour market. For job seekers, the SHAP analysis reveals that years of experience and company location together account for the overwhelming majority of salary variation, which means that geographic mobility and accumulated experience are the two highest leverage strategies for compensation growth in the AI sector [7], [9]. For recruiters and human resource departments, the high coefficient of determination of 0.88 to 0.89 achieved by tree-based ensembles makes the resulting model directly useful as an offer benchmarking tool, particularly when calibrated against the predicted MAE of approximately 15,500 USD which represents a tight error band relative to the median salary of approximately 107,000 USD on this dataset. For policy-makers, the stability of the dominant predictors across three independent interpretability methods provides robust evidence that geographic wage premiums in the AI sector are large and persistent, which is relevant to debates on remote work taxation and digital nomad regulation [2]. For machine learning researchers, the present study illustrates that a relatively simple supervised learning workflow can deliver state of the art performance on a labour market dataset, which lowers the barrier to entry for compensation analytics research and invites further comparative studies on adjacent datasets. For educators, the published Jupyter notebooks provide a turnkey teaching resource for a graduate level course on applied supervised learning, with each notebook corresponding to a single self contained research question that can be used as a class exercise. Finally, the public availability of the entire pipeline through a single GitHub repository, combined with the modest compute requirements, supports the broader open science movement that aims to make machine learning research accessible to under resourced institutions [19]. We hope that the present study will serve both as a reference benchmark for future work in the AI compensation domain and as a template for the kind of multi question, methodologically rigorous studies that the salary prediction literature has historically lacked.

B. *Future Directions*

Several promising future directions follow naturally from the present study. A first direction is the inclusion of richer textual features from the job description and required skills fields, processed by modern language models, which could capture nuance that the current binary skill encoding cannot [4], [27]. A second direction is the explicit modelling of geographic wage adjustment through hierarchical or mixed effects models, which would respect the strong country level grouping observed in the SHAP analysis. A third direction is fairness auditing, in which the trained model would be probed for disparate predictions across protected attributes such as gender, age, or country of residence [9]. A fourth direction is uncertainty quantification through quantile regression or conformal prediction, which would equip the salary estimator with calibrated prediction intervals rather than point estimates only. A fifth direction is temporal extension, namely tracking how the dominant predictors and model accuracies evolve as new editions of the dataset are released year after year, which would allow longitudinal study of compensation dynamics in the AI sector [2], [7]. A final direction is the deployment of the recommended Decision Tree model as an interactive web tool for prospective candidates, accompanied by SHAP based explanations of each individual prediction.

V. Conclusion

This study presented a complete supervised learning benchmark for predicting global AI job salaries using the recent 15,000 record Kaggle Global AI Job Market 2025 dataset. Nine regression models spanning linear, tree-based, ensemble, and kernel families were evaluated under an identical preprocessing pipeline and an identical 80/20 train-test split with a fixed random seed of 42. The headline result is that tree-based ensembles, namely Gradient Boosting, LightGBM, XGBoost, and Random Forest, attain coefficient of determination values between 0.88 and 0.89, comfortably outperforming linear models with values around 0.63 and Support Vector Regression with a value of 0.31. A six way preprocessing ablation confirmed that tree-based models are largely insensitive to scaling and encoding choices, while showing only marginal sensitivity to log transformation of the target. Interpretability analysis using built-in Random Forest importance, permutation importance, and SHAP values consistently identified years of experience and company location as the two dominant predictors, providing a stable and policy relevant insight. Robustness analysis across multiple split ratios, three different cross validation regimes, and six levels of additive Gaussian noise confirmed graceful degradation rather than catastrophic failure. Metric sensitivity analysis showed that the top tier of ensemble models is stable across MAE, RMSE, R squared, MAPE, and MdAE. A final decision matrix selected Decision Tree as the practical recommendation, balancing near top accuracy of 0.87 with substantially higher interpretability, much faster training, and trivial deployment as a single tree. For organisations that prize raw accuracy above all else, Random Forest is the secondary recommendation, while Linear Regression remains the fallback when full coefficient level transparency is mandated. Together these contributions provide a reproducible reference benchmark and a defensible deployment recommendation for the rapidly evolving AI compensation domain.

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